

DiffEEG: A Denoising-Diffusion Foundation Model for Transferable EEG Representations Across Seizure, Abnormality, Motor, and Sleep Tasks

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Abstract—Electroencephalography (EEG) foundation models aim to learn task- and subject-invariant representations that transfer to many downstream problems. Most existing models are pretrained with masked or contrastive objectives and are evaluated with full fine-tuning on benchmarks whose preprocessing matches pretraining. We present DIFEEG, a 40.7M-parameter denoising-diffusion model that learns generic EEG representations by reconstructing clean signals from noisy inputs, and that is used as a frozen feature extractor for downstream tasks. Representations are obtained by probing the diffusion denoiser at five noise levels and pooling its hierarchical encoder activations into a 4800-dimensional descriptor. A lightweight channel-harmonization front end projects any electrode montage onto a canonical 22-channel frame, allowing a single fixed-input model to process datasets recorded with different hardware. We adapt the embedding with a linear probe, partial fine-tuning, and a reinforcement-learning (RL) head that directly optimizes macro-F1. Across seven datasets spanning four task families—seizure detection (TUSZ, Siena, Bonn), EEG abnormality and event classification (TUAB, TUEV), motor imagery (EEGMMIDB), and sleep staging (Sleep-EDFx)—the diffusion embedding transfers effectively: it reaches 0.993 ROC-AUC on Bonn seizure detection, 94.0% accuracy on patient-wise TUSZ seizure detection, 0.878 ROC-AUC (79.3% accuracy) on TUAB abnormality detection, and above-chance accuracy on tasks the backbone never saw during pretraining, including motor imagery (61.5%) and six-class sleep staging (53.9%). The RL head is particularly effective under extreme class imbalance, improving minority-class precision by up to eight-fold on the 0.64%-prevalence Siena corpus—the operating regime of continuous clinical monitoring. Compared with recent foundation models on accuracy, macro/weighted-F1, and ROC-AUC, DIFEEG is competitive on clinical abnormality detection while offering broad, low-cost transfer from a single generative backbone. To support clinical interpretability, we analyze the learned representation: t-SNE of the frozen embedding shows that pretraining alone organizes clinically meaningful event categories, and gradient-based saliency localizes the scalp regions and time segments that drive epileptiform decisions, in agreement with known electrophysiology.

Index Terms—EEG, foundation model, denoising diffusion, representation learning, transfer learning, reinforcement learning, seizure detection, sleep staging.

I. INTRODUCTION

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ELECTROENCEPHALOGRAPHY (EEG) is the primary non-invasive tool for diagnosing epilepsy and screening for cerebral abnormality, and it underpins a wide range of clinical and brain-computer interface (BCI) applications, including seizure detection, abnormality screening, sleep staging, and motor decoding [1], [2]. Manual EEG review is labor-intensive, requires scarce sub-specialist expertise, and suffers from substantial inter-rater variability; automated decision support could therefore materially improve the consistency and reach of neurological care. Yet clinical EEG poses challenges that generic deep learning handles poorly: pathological events (seizures, epileptiform discharges) are rare, occupying a small fraction of continuous recordings, so models optimized for accuracy collapse onto the majority “normal” class; and models trained for one task, montage, or institution rarely transfer to another, while the volume of expert annotation required to retrain them is a persistent barrier. These constraints have motivated EEG foundation models (FMs): networks pretrained self-supervised on large unlabeled corpora and adapted to specific downstream tasks [3], [4], [5], [6].

Two characteristics dominate the current FM literature. First, nearly all models are pretrained with masked or contrastive objectives adapted from language and vision. Second, they are usually evaluated by full fine-tuning on benchmarks whose channel montage, sampling rate, and window length match those used during pretraining. As a result, it is not well understood whether a generative pretraining objective produces representations as transferable as masked prediction, nor how well a pretrained EEG model behaves as a frozen embedding that must adapt to genuinely different montages and tasks with little fine-tuning—the regime that matters most for rapid BCI calibration and low-label clinical deployment.

This paper studies both questions using DIFEEG, a denoising-diffusion model that is pretrained purely to reconstruct clean EEG and then used as a general-purpose feature extractor. Rather than training a task-specific network end-to-end, we probe the frozen diffusion denoiser at several noise levels and pool its multi-scale encoder activations into a single descriptor. Because the backbone accepts a fixed 22-channel input, we introduce a channel-harmonization front end that maps any montage onto the model’s canonical frame, enabling a single frozen model to be exercised on datasets with 1–64 native channels and 100–512 Hz native sampling rates. The embedding is then adapted with a head-only classifier, partial fine-tuning, or a reinforcement-learning head that directly optimizes macro-F1.

The contributions of this work are:

- 1) A multi-timestep diffusion-probing embedding that turns a generative EEG denoiser into a compact 4800-dimensional feature extractor supporting head-only classification.
- 2) A channel-harmonization pipeline that projects arbitrary montages onto a canonical 22-channel frame, allowing a fixed-input model to be applied to heterogeneous datasets.
- 3) A broad transfer evaluation across seven datasets and four task families, with head-only adaptation, fine-tuning, and reinforcement-learning adaptation, and a direct comparison against recent foundation models on accuracy, macro/weighted-F1, ROC-AUC, and AUC-PR.
- 4) An interpretability analysis tailored to clinical use: embedding-space visualization showing that the pre-trained representation organizes clinically meaningful event categories, and gradient-based saliency that localizes the electrodes and time segments driving epileptiform decisions.

II. RELATED WORK

A. EEG foundation models

BIOT [3] tokenizes biosignals into fixed-length segments to learn across datasets with mismatched channels. LaBraM [4] introduces a neural tokenizer with vector-quantized spectral prediction and scales to hundreds of millions of parameters. CBraMod [5] uses a criss-cross transformer that models temporal and spatial dependencies separately and popularized linear-probing evaluation. EEGPT [6] aligns spatio-temporal representations with a dual self-supervised objective. More recent models target montage generalization: NEURIPT [13] adds 3D electrode coordinate encodings, amplitude-aware masked pretraining, and a progressive mixture-of-experts; REVE [14] scales masked autoencoding to 60 000 hours from 92 datasets and 25 000 subjects with a 4D positional encoding that natively accepts arbitrary montages. In the clinical seizure domain, BIOSERENITY-E1 [15] combines spectral tokenization with masked prediction and reports strong AUROC on the TUH Seizure corpus.

B. Diffusion models for representation learning

Denoising diffusion probabilistic models [8], [9] learn to invert a gradual noising process and are the dominant paradigm for generative modeling. Beyond synthesis, the intermediate activations of a trained denoiser encode semantically meaningful features whose abstraction level varies with the noise level. DIFFEEG [7] first applied this idea to seizure detection. The present work differs in two ways: it studies the model as a general embedding rather than a seizure classifier, and it evaluates transfer across four task families under montage and task shift rather than a single in-domain benchmark.

III. METHODS

A. Diffusion backbone

DIFFEEG is a conditional 1D U-Net trained to reverse a cosine-schedule Gaussian noising process over $T = 1000$

timesteps [8], [9]. Given a clean segment $x_0 \in \mathbb{R}^{C \times L}$ with $C = 22$ channels and $L = 1280$ samples (5 s at 256 Hz), the forward process adds noise as

$$x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}), \quad (1)$$

and the network $\epsilon_\theta(x_t, t)$ is trained with the noise-prediction loss $\mathcal{L} = \mathbb{E}_{x_0, \epsilon, t} \|\epsilon - \epsilon_\theta(x_t, t)\|^2$. The backbone uses base width $m = 64$, channel multipliers $[1, 2, 4, 8]$ (encoder feature widths $\{64, 128, 256, 512\}$), three residual blocks per level with feature-wise linear modulation (FiLM) [10] conditioning on the timestep, and multi-head self-attention at the deeper encoder levels and the bottleneck. The full model has 40 679 574 parameters. The encoder-decoder is trained end-to-end during pretraining; for downstream use the decoder is discarded and only the encoder is retained as a feature extractor.

B. Network architecture in detail

Figure 1 details the backbone used throughout this work (the training configuration referred to as V2, with base width $m = 64$, channel multipliers $[1, 2, 4, 8]$, three residual blocks per encoder level, and time embedding dimension 768). The network takes the noisy input $x_t \in \mathbb{R}^{22 \times 1280}$ together with the scalar timestep t and outputs the predicted noise $\hat{\epsilon} \in \mathbb{R}^{22 \times 1280}$.

The timestep is first mapped to a 768-dimensional embedding by a sinusoidal position encoding followed by a two-layer MLP ($768 \rightarrow 1536 \rightarrow 768$, SiLU activation); this embedding conditions every residual block in the network. A 3×3 convolution (`init_conv`) projects the 22 input channels to the base width of 64.

The encoder has four resolution levels with channel widths $\{64, 128, 256, 512\}$. Each level applies three residual blocks at a fixed channel count and temporal length, followed by a strided (stride = 2) 3×3 convolution that halves the sequence length; three such downsampling steps take the sequence from 1280 samples to 160 samples by the end of the encoder. Self-attention (8 heads) is applied only in the last residual block of the two deepest encoder levels (256 and 512 channels), leaving the shallow, high-resolution levels purely convolutional for efficiency. The bottleneck operates at 160 samples and 512 channels with three residual blocks, all with self-attention.

The decoder mirrors the encoder in reverse, with four levels returning the channel width to $\{512, 256, 128, 64\}$ and the sequence length from 160 back to 1280 samples via three upsampling steps (linear interpolation by a factor of 2 followed by a 3×3 convolution). Because the additive skip connection is folded into the first residual block at each decoder level, the decoder uses one more residual block per level than the encoder, four instead of three. Self-attention is applied in the first residual block of the two decoder levels closest to the bottleneck (512 and 256 channels). Each decoder level receives an additive skip connection from the output of the corresponding encoder level (resized by linear interpolation if the temporal lengths differ after non-integer downsampling), which is added to the input of that level's first residual block, in the same spirit as the original U-Net [12]. A final instance normalization, a 1×1 convolution back to 64 channels, and a 3×3 convolution to 22 channels produce the predicted noise.

Every residual block (Fig. 2) follows the same pattern: instance normalization, SiLU, a 3×3 convolution, feature-wise linear modulation (FiLM) [10] that applies a per-channel scale and shift derived from the timestep embedding, a second normalization–SiLU–dropout–convolution pair, an optional multi-head self-attention over the temporal axis, and a residual connection (with a 1×1 projection when the channel count changes). In total the network contains 31 residual blocks (12 in the encoder, 3 in the bottleneck, and 16 in the decoder), of which 7 include the attention module, for 40 679 574 parameters overall.

C. Multi-timestep diffusion-probing embedding

To turn the denoiser into a feature extractor we probe the encoder at a fixed set of diffusion timesteps

$$\mathcal{T} = \{50, 250, 500, 750, 950\} \quad (2)$$

that uniformly span the noise schedule. Low-noise probes preserve fine temporal detail (e.g. spikes, spindles) while high-noise probes capture coarse morphology. For an input segment x , we form the noised input x_t at each $t \in \mathcal{T}$, run the encoder, and apply global adaptive average pooling to each of the four encoder levels $\ell \in \{1, 2, 3, 4\}$:

$$f_\ell^{(t)} = \frac{1}{L_\ell} \sum_{i=1}^{L_\ell} h_{\ell,i}^{(t)} \in \mathbb{R}^{d_\ell}, \quad d_\ell \in \{64, 128, 256, 512\}. \quad (3)$$

Concatenating across all levels and timesteps yields the embedding

$$z = \bigoplus_{t \in \mathcal{T}} \bigoplus_{\ell=1}^4 f_\ell^{(t)} \in \mathbb{R}^D, \quad D = |\mathcal{T}| \times \sum_{\ell} d_\ell = 5 \times 960 = 4800. \quad (4)$$

This 4800-dimensional vector is the fixed EEG descriptor used by all downstream tasks. Because it is computed from a frozen backbone with deterministic noising seeds, it is precomputed and cached once per dataset. Fig. 3 summarizes the full pipeline.

D. Head-only

The default adaptation is head-only training: the backbone is frozen and a two-layer multilayer perceptron (MLP) classifier with batch normalization, GELU activation, and dropout is trained on the cached embedding z . Training uses weighted cross-entropy with inverse-frequency class weights $w_k = N/(K n_k)$, where N is the total number of samples, K the number of classes, and n_k the count of class k , to counter class imbalance. Head-only training isolates the quality of the pretrained representation, since only the head is optimized.

E. Embedding-only evaluation

A natural concern is that the 2-layer MLP head is itself a small trainable classifier, so its performance mixes two confounded signals: (i) the quality of the embedding, and (ii) the additional capacity of the head. To isolate (i), we report a strictly non-parametric evaluation of the embedding using k-nearest-neighbor classification on the 4800-d embedding

space. We adopt k-NN rather than the more common single-layer logistic-regression classifier because k-NN is bounded above by the best parametric probe yet makes no assumption about the form of the decision boundary. In our setting this is the more useful endpoint: it directly measures the local class consistency of the embedding space, which is the property a downstream user of the embedding most cares about. When additional capacity is warranted, we unfreeze part or all of the backbone. A gentle differential learning rate is used: the head is trained at $\eta_{\text{head}} = 5 \times 10^{-4}$ while unfrozen backbone layers use $\eta_{\text{backbone}} = 0.01 \eta_{\text{head}}$, which avoids the optimization collapse that a single aggressive learning rate causes on small downstream datasets. Unfreezing is controlled from `last_level` (only the deepest encoder level) to `all` (the full encoder and bottleneck).

F. Reinforcement-learning decision head

For imbalanced tasks, per-sample cross-entropy tends to favor the majority class. We therefore add a reinforcement-learning (RL) decision head that directly optimizes the target evaluation metric. A policy network $\pi_\phi : \mathbb{R}^D \rightarrow \mathbb{R}^K$ produces class logits; during training we sample predictions from the resulting Categorical distribution and receive the batch macro-F1 as the reward R . Using the REINFORCE estimator [11] with an exponential-moving-average baseline b for variance reduction, the policy-gradient loss is

$$\mathcal{L}_{\text{PG}} = -(R - b) \frac{1}{N_b} \sum_{i=1}^{N_b} \log \pi_\phi(\hat{y}_i | z_i), \quad (5)$$

and the total objective combines it with weighted cross-entropy,

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{PG}}, \quad \lambda = 0.1. \quad (6)$$

The macro-F1 reward makes the head sensitive to rare classes and generalizes to any number of classes, so the same mechanism is used for binary and multi-class tasks.

G. Channel harmonization

The backbone accepts exactly 22 channels in a fixed referential order (FP1, FP2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, A1, A2, FZ, CZ, PZ, ROC). For each recording we (i) canonicalize electrode names, (ii) map them onto the 22-slot frame by exact match and then a curated alias table (e.g. T7 $\rightarrow$ T3, P7 $\rightarrow$ T5, TP9 $\rightarrow$ A1; for bipolar-only montages Fpz–Cz $\rightarrow$ FZ, Pz–Oz $\rightarrow$ PZ), (iii) zero-fill any target channel with no matching electrode, and (iv) resample the session to 256 Hz before windowing. Unmatched channels are zero-filled rather than interpolated, so the model never receives fabricated signal. The number of real channels a dataset contributes is therefore an explicit quantity (Table I).

IV. EXPERIMENTS AND IMPLEMENTATION

A. Data processing

All datasets are converted to the model’s input format—22 channels, 256 Hz, 5 s (1280-sample) windows—using the

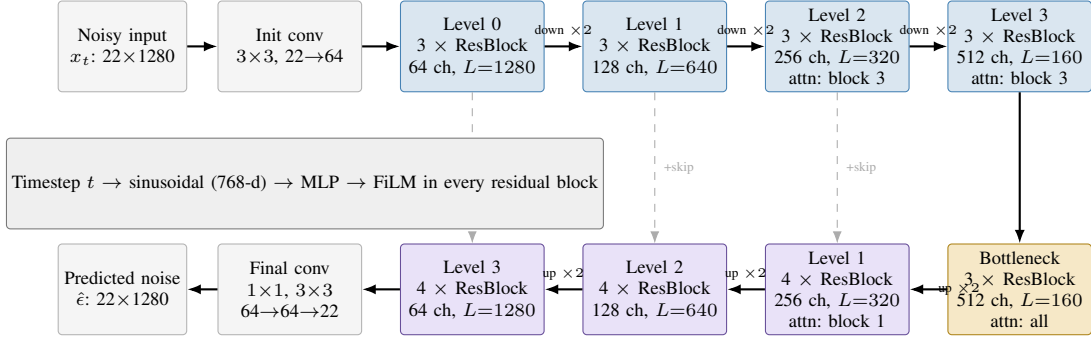


Fig. 1. Detailed architecture of the DiffEEG 1D U-Net (V2 configuration). Top row: encoder, four levels with channel widths 64, 128, 256, 512 and three residual blocks per level; strided convolutions halve the sequence length between levels. Bottom row: decoder, mirroring the encoder with one extra residual block per level (four) to absorb the additive skip connection, and transposed-resolution upsampling. Dashed arrows denote the additive skip connections from each encoder level into the corresponding decoder level. Self-attention (8 heads) is used only in the deepest two encoder levels, the full bottleneck, and the two decoder levels nearest the bottleneck.

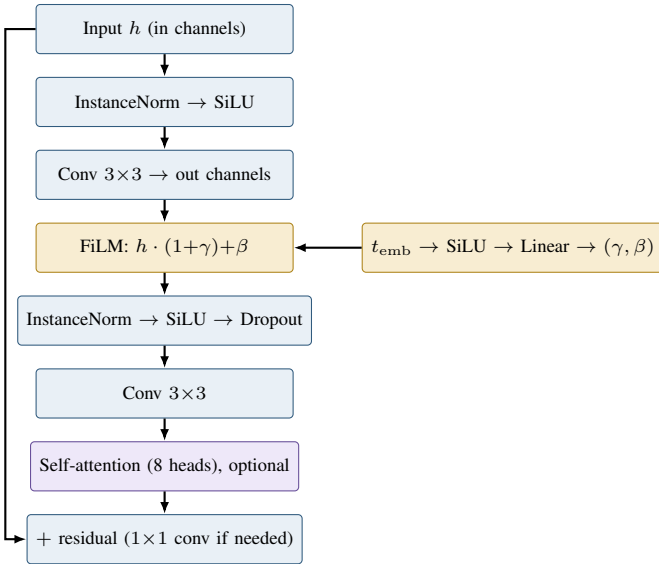


Fig. 2. Internal structure of a single residual block. The timestep embedding t_{emb} is projected to a per-channel scale γ and shift β (FiLM) that condition the block on the diffusion noise level. The self-attention step is present only in 7 of the 31 blocks in the network (Fig. 1).

harmonization pipeline of Section III-G. Table I summarizes each dataset’s native characteristics and the outcome of harmonization. For datasets whose raw amplitude scale differs from the pretraining convention (e.g. Siena, recorded in volts), we compute dataset-specific normalization statistics; for datasets without global statistics we apply per-window z-scoring. TUEV carries short discrete event annotations rather than continuous labels, so windows are extracted centered on each merged event rather than on a fixed grid.

B. Pretraining

The diffusion backbone is pretrained, fully self-supervised, on $\approx 1.45\text{M}$ unlabeled five-second segments of clinical EEG drawn from three corpora: the non-seizure segments of the Temple University Hospital Seizure Corpus (TUSZ) and of CHB-MIT, together with the normal recordings of the Temple University Hospital Abnormal Corpus (TUAB). Segments are

extracted with 30% overlap, notch-filtered at 50 Hz, band-pass filtered between 1 and 75 Hz with a zero-phase Butterworth filter, and channel-wise normalized. Training minimizes the noise-prediction loss with a cosine variance schedule ($T = 1000$) using AdamW; no labels are used.

We note one caveat that affects interpretation of the TUAB result: because the *normal* recordings of TUAB—including its official held-out test-normal split—are part of this pretraining corpus, the downstream TUAB abnormality-detection number (Table V) is not a strict out-of-corpus transfer result. The backbone has already seen the TUAB normal distribution during self-supervised pretraining, although it never observed any abnormal recording or any label. TUAB should therefore be read as an in-domain evaluation; the seizure (Siena, Bonn), motor-imagery, sleep, emotion, and MDD corpora, which are entirely absent from pretraining, are the genuine transfer tests.

C. Fine-tuning and adaptation

Downstream heads are trained with AdamW and a cosine-annealing schedule. Head-only training uses $\eta = 1 \times 10^{-3}$ and early stopping on validation macro-F1; fine-tuning uses the differential learning rate of Section III-D; the RL head uses $\lambda = 0.1$, a baseline momentum of 0.9, and a macro-F1 reward. Unless otherwise noted, class imbalance is handled with inverse-frequency weighted cross-entropy.

D. Implementation details

All experiments use PyTorch on NVIDIA A100 GPUs. The frozen embedding is cached once per dataset to amortize the cost of multi-timestep probing. Evaluation metrics are chosen per task: for imbalanced binary tasks we report ROC-AUC, PR-AUC, and minority-class F1; for multi-class tasks we report accuracy and macro-F1. Weighted-average F1 is reported only where required for comparison with prior work.

V. RESULTS

Fig. 4 visualizes the best score obtained on each dataset against its chance level, and Tables II–VIII report the full adaptation ladder per dataset; the following subsections discuss each task family.

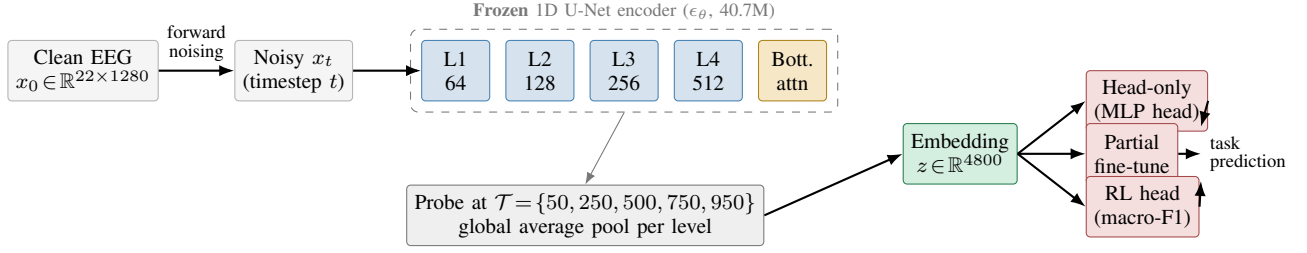


Fig. 3. DiffEEG pipeline. A 1D U-Net is pretrained to denoise EEG (generative objective). For downstream use the frozen encoder is probed at five diffusion timesteps; per-level global average pooling and concatenation yield a 4800-dimensional embedding, which is adapted by a head-only classifier, partial fine-tuning, or a reinforcement-learning head.

TABLE I
DATASETS, NATIVE ACQUISITION CHARACTERISTICS, AND THE RESULT OF HARMONIZATION ONTO THE CANONICAL 22-CHANNEL FRAME.

Dataset	Task family	Native chans	Native rate	Real ch. / 22	Zero-filled	Split
TUSZ	Seizure detection	29	256 Hz	22	0	patient-wise (official eval)
Siena	Seizure detection	45	512 Hz	19	3	patient-wise (11/3)
Bonn	Seizure detection	1	173.6 Hz	1	21	record-wise
TUAB	Abnormality detection	36	250 Hz	22	0	official train/eval
TUEV	Event classification	27	250 Hz	22	0	official train/eval
EEGMMIDB	Motor imagery	64	160 Hz	19	3	subject-wise
Sleep-EDFx	Sleep staging	2	100 Hz	2	20	subject-wise

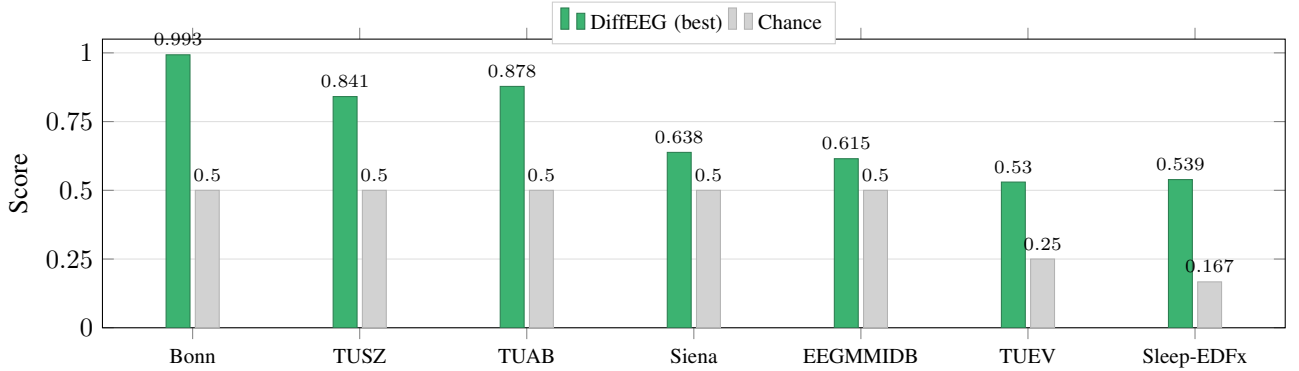


Fig. 4. Best DiffEEG score per dataset versus chance level. Metric per dataset: ROC-AUC for Bonn, TUSZ, and TUAB; macro-F1 for Siena and TUEV; accuracy for EEGMMIDB (2-class) and Sleep-EDFx (6-class). Every task exceeds its chance baseline, from strong seizure/abnormality transfer to above-chance motor imagery and sleep staging.

Each dataset is reported in its own table listing every adaptation regime we evaluated—trained head only (frozen backbone), and fine-tuning with the backbone unfrozen, each with and without the RL head. Bold marks the best-accuracy row; “—” denotes a configuration that was not run. Binary scores are reported at the F1-optimal operating point (threshold tuned on the validation set); ROC-AUC and PR-AUC are threshold-free.

A. Seizure detection: TUSZ, Siena, Bonn

On the official patient-wise TUSZ evaluation split (228 667 segments, 6.7% ictal prevalence, Table II), the trained head already reaches accuracy 0.908 and ROC-AUC 0.850. Un-freezing the backbone with the RL head gives the best operating point overall: accuracy 0.940, macro-F1 0.732, seizure-F1 0.496, ROC-AUC 0.841, and PR-AUC 0.484—a substantial gain in sensitivity to the rare ictal class. A four-class

TABLE II
TUSZ SEIZURE DETECTION (PATIENT-WISE EVALUATION SPLIT). SZ-F1 = SEIZURE-CLASS F1.

Adaptation	Acc	Macro-F1	Sz-F1	ROC-AUC	PR-AUC
Head only, no RL	0.908	0.697	0.444	0.850	0.429
Head only, +RL	0.911	0.689	0.426	0.836	0.369
Unfrozen, no RL	—	—	— (not run)	—	—
Unfrozen, +RL	0.940	0.732	0.496	0.841	0.484

seizure-subtype task under strict patient-wise cross-validation additionally reaches macro-F1 0.363 ± 0.020 , with common subtypes separating well and rare subtypes remaining difficult.

Siena is an extreme-imbalance benchmark (0.64% ictal prevalence, Table III). A frozen trained head reaches accuracy 0.951 and macro-F1 0.528 but calls many false positives (seizure precision 0.046). Full fine-tuning with the RL head

TABLE III
SIENA SEIZURE DETECTION (0.64% ICTAL PREVALENCE). SZ-P = SEIZURE PRECISION.

Adaptation	Acc	Macro-F1	Sz-F1	Sz-P	ROC-AUC
Head only, no RL	0.951	0.528	0.080	0.046	0.734
Head only, +RL			– (not run)	–	
Unfrozen, no RL			– (not run)	–	
Unfrozen, +RL	0.992	0.638	0.280	0.366	–

TABLE IV
BONN SEIZURE VS. NORMAL (SINGLE ACTIVE CHANNEL).

Adaptation	Acc	Macro-F1	W-F1	ROC-AUC	PR-AUC
Zero-shot	0.846	0.643	–	0.716	0.647
Head only, no RL	0.930	0.903	0.934	0.993	0.974
Unfrozen			– (not run)	–	

raises accuracy to 0.992 and macro-F1 to 0.638, lifting seizure precision roughly eight-fold (to 0.366) and reducing false positives by an order of magnitude—a far more usable operating point for continuous monitoring. Zero-shot reuse of the TUSZ head is unusable here (ROC-AUC 0.626, tens of thousands of false positives).

Bonn is a small curated seizure/normal benchmark in which each record provides a single trace placed in one of the 22 channels (Table IV). Even so, the trained head reaches ROC-AUC 0.993, PR-AUC 0.974, and accuracy 0.930; zero-shot reuse of the TUSZ head already gives ROC-AUC 0.716. Seizure morphology is thus strongly captured even from minimal spatial coverage.

B. EEG abnormality and events: TUAB, TUEV

On the roughly balanced TUAB abnormality-detection task (Table V), a frozen trained head reaches accuracy 0.645 and ROC-AUC 0.776. Unfreezing the backbone improves this markedly: the best-accuracy configuration reaches accuracy 0.793, F1 0.771, ROC-AUC 0.878, and PR-AUC 0.877.

TUEV is a fine-grained event-classification corpus. The three epileptiform families (spike/sharp-wave, generalized and periodic epileptiform discharges) are mutually confused at the model’s temporal resolution; the original six-class task therefore collapses to accuracy 0.336. Merging the three into a single epileptiform class gives a four-class task (artifact, background, epileptiform, eye movement) on which full fine-tuning reaches accuracy 0.628 and macro-F1 0.530, with epileptiform precision 0.894 (Table VI); adding the RL head is comparable (0.606 / 0.499). The dominant errors are between the two non-epileptiform background categories, not on the clinically important epileptiform class.

C. Motor imagery and sleep staging: EEGMMIDB, Sleep-EDFx

EEGMMIDB is a two-class imagined-movement task (left vs. right fist) far outside the pretraining domain (Table VII). The frozen trained head reaches 53.2% accuracy, and progressively unfreezing the backbone monotonically improves this to 57.2% and 61.5% (macro-F1 0.615), well above the 50%

TABLE V
TUAB ABNORMALITY DETECTION (BALANCED BINARY). n = NUMBER OF UNFROZEN DOWNSAMPLING BLOCKS; SENS/SPEC = SENSITIVITY/SPECIFICITY.

Adaptation	Acc	F1	Sens	Spec	ROC-AUC	PR-AUC
Head only, no RL	0.645	0.690	0.855	0.466	0.776	0.756
Unfrozen ($n=2$), no RL	0.793	0.771	0.753	0.828	0.878	0.877
Unfrozen ($n=3$), no RL	0.789	0.772	0.776	0.800	0.862	0.854
+RL variants			– (not run)	–		

TABLE VI
TUEV EVENT CLASSIFICATION. ROWS ARE THE FOUR-CLASS MERGED TASK UNLESS MARKED (6-CL).

Adaptation	Acc	Macro-F1	W-F1
Head only, no RL	0.560	0.445	0.588
Unfrozen, no RL	0.628	0.530	0.651
Unfrozen, +RL	0.606	0.499	0.632
Unfrozen, +RL (6-cl)	0.336	0.321	0.347

chance. A pooled-random-split diagnostic (mixing subjects) yields a nearly identical frozen result (56.9%), indicating that the limitation is the pretraining domain rather than cross-subject transfer.

Sleep-EDFx is a six-class staging task in which only two bipolar EEG channels are available (Table VIII). Despite this minimal spatial coverage, the embedding reaches 53.9% accuracy and macro-F1 0.539, far above the 16.7% chance. Per class, wake and deep sleep (N4) separate best (F1 0.831 and 0.683), while the light-sleep transitions N1/N2 are the weakest (F1 0.418/0.353), matching their known difficulty for human scorers. RL adds little here because the task is class-balanced by design.

D. Mental disorder detection: Mumtaz2016

Mumtaz2016 is a binary major-depressive-disorder (MDD) versus healthy-control task from 64 subjects (34 MDD, 30 control) recorded on the 19-electrode 10–20 system, which map onto 19 of our 22 canonical channels. We use a subject-independent split (no subject appears in both train and test). To place our result alongside the published Mumtaz2016 leaderboard, we reprocess the data following the protocol adopted by CBraMod and REVE [5], [14]: only the eyes-open and eyes-closed resting sessions (the P300 TASK session dropped), a 50 Hz notch filter, a band-pass filter, and a common-average reference. We keep the backbone’s native 256 Hz (their pipeline resamples to 200 Hz) and use a 0.1–30 Hz band. For contrast we also report our default referential setup, which additionally includes the TASK session and applies no band-pass.

The preprocessing choice dominates the result (Table IX). Under the matched pipeline a frozen trained head reaches balanced accuracy 0.812, ROC-AUC 0.935, and PR-AUC 0.942—a 0.10 ROC-AUC gain over our default referential preprocessing (0.832), recovered almost entirely by the average reference and band-pass. Notably, under the matched pipeline the frozen head is the strongest configuration: full fine-tuning lowers balanced accuracy to 0.767, overfitting the small

TABLE VII
EEGMMIDB TWO-CLASS MOTOR IMAGERY (SUBJECT-WISE SPLIT).

Adaptation	Acc	Macro-F1	W-F1	ROC-AUC
Head only, no RL	0.532	0.521	0.523	0.586
Unfrozen (last level), no RL	0.572	0.572	0.572	–
Unfrozen (all), no RL	0.615	0.615	0.615	–
+RL variants	– (not run) –			

TABLE VIII
SLEEP-EDFX SIX-CLASS SLEEP STAGING (TWO BIPOLAR CHANNELS).

Adaptation	Acc	Macro-F1	W-F1
Head only, no RL	0.535	0.530	0.530
Unfrozen, no RL	0.535	0.538	0.538
Unfrozen, +RL	0.539	0.539	0.539

(≈ 5600 -window) training set—consistent with the linear-probe strength reported for other EEG foundation models. Our best balanced accuracy (0.812) remains below the leaderboard (EEGNet 0.923, CBraMod 0.956, REVE-Base 0.964), but the AUC metrics (ROC-AUC 0.935, PR-AUC 0.942) approach the strongest task-specific baseline (EEGNet AUROC 0.964); the balanced-accuracy gap is partly a decision-threshold effect, as the frozen head is MDD-biased at the default threshold (control recall 0.66 versus MDD 0.96).

E. Linear probing: the frozen embedding across tasks

Table X consolidates the linear-probing result—the frozen DiffEEG embedding with only a trained classifier head and no backbone fine-tuning—across all eight downstream datasets, next to the best score reached anywhere in the adaptation ladder. The probe classifier is the lightweight two-layer MLP head described in Section III-D ($4800 \rightarrow 512 \rightarrow 256 \rightarrow K$, with batch normalization, GELU, and dropout), trained with class-weighted cross-entropy while every backbone weight stays frozen; we use “linear probing” in the common sense of fitting only a small head on top of the fixed representation. As a stricter, parameter-free alternative we also evaluate the same frozen embedding with a k -nearest-neighbor classifier (Section III-E)—a classifier with no trained weights at all (Table XI). The frozen embedding alone is already highly discriminative for morphology-driven clinical tasks: without touching the backbone it reaches ROC-AUC 0.993 (Bonn), 0.935 (Mumtaz MDD), and 0.850 (TUSZ), and on TUSZ, Mumtaz, and Bonn the frozen probe is in fact the strongest configuration. Fine-tuning helps most where the task is furthest from the pretraining domain (EEGMMIDB motor imagery $0.532 \rightarrow 0.615$; TUAB $0.776 \rightarrow 0.878$ ROC-AUC) and adds little on the tasks the frozen embedding already solves.

The parameter-free k NN probe (Table XI) sharpens this picture: with no trained weights at all, the embedding’s local neighborhoods already separate the morphology-driven clinical tasks well above chance—Mumtaz MDD 0.791, Bonn 0.775, Siena 0.615 (chance 0.5), and TUEV 0.328 (chance 0.25)—but sit exactly at chance for the out-of-domain BCI and sleep tasks (EEGMMIDB 0.500, Sleep-EDFx 0.167), which only become separable once a head is trained or the backbone

TABLE IX
MUMTAZ2016 MDD DETECTION (BINARY, SUBJECT-INDEPENDENT SPLIT). TWO PREPROCESSING SETUPS: OUR DEFAULT REFERENTIAL MONTAGE (EC+EO+TASK, NO BAND-PASS) AND THE CBraMod/REVE-MATCHED PIPELINE (EC+EO ONLY, 50 Hz NOTCH, BAND-PASS, COMMON-AVERAGE REFERENCE). BAL-ACC = BALANCED ACCURACY; ROC-AUC/PR-AUC ARE REPORTED FOR THE FROZEN-HEAD RUNS.

Adaptation	Bal-Acc	Macro-F1	ROC-AUC	PR-AUC
<i>Default preprocessing (EC+EO+TASK, referential)</i>				
Head only, no RL	0.756	0.758	0.832	0.811
Unfrozen, no RL	0.768	0.769	–	–
Unfrozen, +RL	0.763	0.764	–	–
<i>CBraMod/REVE-matched (EC+EO, notch + band-pass + avg-ref)</i>				
Head only, no RL	0.812	0.818	0.935	0.942
Unfrozen, no RL	0.767	0.770	–	–
Unfrozen, +RL	0.768	0.772	–	–

TABLE X
LINEAR PROBING ACROSS TASKS: FROZEN DIFFEEG EMBEDDING + TRAINED HEAD (NO BACKBONE FINE-TUNING). “BEST” IS THE TOP OF THE FULL ADAPTATION LADDER FOR THE SAME METRIC (TABLES II–IX). METRIC: R = ROC-AUC, MF1 = MACRO-F1, ACC = ACCURACY.

Dataset	Task	Metric	Linear probe	Best
Bonn	Seizure	R	0.993	0.993
TUSZ	Seizure	R	0.850	0.841
Siena	Seizure	mF1	0.528	0.638
TUAB	Abnormality	R	0.776	0.878
TUEV	Events (4-cl)	mF1	0.445	0.530
EEGMMIDB	Motor imagery	Acc	0.532	0.615
Sleep-EDFx	Sleep (6-cl)	Acc	0.535	0.539
Mumtaz2016	MDD	R	0.935	0.935

is fine-tuned. The pretraining domain is thus imprinted directly on the geometry of the embedding space. True zero-shot transfer (reusing the TUSZ seizure head with no training at all) is meaningful only within the seizure family and is reported per dataset above (Bonn ROC-AUC 0.716; Siena 0.626).

VI. MODEL INTERPRETATION AND EMBEDDING ANALYSIS

Clinical adoption of automated EEG analysis requires not only accuracy but also representations that are inspectable and physiologically plausible. We therefore analyze what the diffusion embedding encodes and which signal features drive its clinical decisions, using the frozen pretrained backbone and the finetuned TUEV classifier.

A. The embedding organizes clinical categories

Fig. 5 shows a two-dimensional t-SNE projection of the 4800-dimensional embedding. For Bonn (seizure vs. normal), the frozen pretrained embedding, computed with no task training whatsoever, separates the two classes into clearly distinct regions, and a k -nearest-neighbor classifier on the raw embedding reaches 77.5% balanced accuracy (chance 50%). This confirms that seizure morphology is captured by pretraining alone, consistent with the strong head-only result on Bonn (Table IV). For Siena, an independent seizure-detection corpus with extreme class imbalance (Table III), the frozen embedding already separates several sub-regions of seizure windows from the normal majority, and full finetuning

TABLE XI

PARAMETER-FREE (NON-PARAMETRIC) PROBE OF THE FROZEN 4800-DIMENSIONAL DIFFEEG EMBEDDING: k -NEAREST-NEIGHBOR CLASSIFICATION ($k=10$, DISTANCE-WEIGHTED), SCORED BY 5-FOLD CROSS-VALIDATED BALANCED ACCURACY (IMBALANCE-ROBUST). A CLASSIFIER WITH NO TRAINED WEIGHTS—IT USES THE TARGET LABELS ONLY AS THE NEIGHBOR BANK, AND IS THEREFORE DISTINCT FROM TRUE ZERO-SHOT TRANSFER (REUSING THE TUSZ SEIZURE HEAD, REPORTED PER DATASET). CHANCE = $1/(\# \text{ CLASSES})$. TUAB AND TUSZ USE THE PATIENT-WISE PIPELINE AND ARE EVALUATED SEPARATELY.

Dataset	Task	Chance	k NN bal-acc
Bonn	Seizure (2-cl)	0.500	0.775
Mumtaz2016	MDD (2-cl)	0.500	0.791
Siena	Seizure (2-cl)	0.500	0.615
TUEV	Events (4-cl)	0.250	0.328
EEGMMIDB	Motor (2-cl)	0.500	0.500
Sleep-EDFx	Sleep (6-cl)	0.167	0.167

with the RL head visibly consolidates the seizure windows into more coherent clusters, raising k NN balanced accuracy from 61% to 71% (chance 50%). The embedding is thus not a black box but an organized feature space in which clinically distinct categories occupy increasingly distinguishable regions as adaptation proceeds.

B. Saliency localizes the electrophysiology

To identify which inputs drive a decision, we compute gradient-based saliency for the epileptiform class—the mean absolute gradient of the epileptiform logit with respect to the input, averaged over all epileptiform test windows (Fig. 6). The resulting scalp map is dominated by temporal and frontotemporal electrodes (T4, F7, F3, T5) together with frontal FP2, which matches the well-established temporal-lobe predominance of interictal epileptiform discharges; occipital and midline-central channels contribute least. The temporal saliency profile is distributed across the entire 5 s window with transient peaks rather than being concentrated at a single instant, consistent with the intermittent, sharp-transient nature of epileptiform activity. Because the saliency is expressed directly over the standard 10–20 montage, it can be overlaid on a clinician’s electrode map, providing a physiologically interpretable rationale for each decision.

VII. DISCUSSION

The results show that a purely generative diffusion backbone yields a representation that transfers across four distinct task families from a single pretraining corpus. Three patterns are worth highlighting.

a) Transfer tracks task relatedness: Seizure-like tasks transfer best (Bonn ROC-AUC 0.993; strong minority precision on Siena), which is expected given seizure-corpus pretraining. Sleep staging transfers moderately even from two channels, indicating that the encoder captures broadband oscillatory structure (delta/theta power, spindles) that is not seizure specific. Motor imagery transfers the least, and a pooled-split diagnostic (mixing subjects across train and test) barely changes the frozen result, localizing the bottleneck to the pre-training domain—the backbone has not learned sensorimotor-rhythm features—rather than to cross-subject variability.

TABLE XII

WINDOW-SIZE SENSITIVITY. THE BACKBONE PERFORMS BEST NEAR ITS 5 s PRETRAINING LENGTH; EXTREME DEVIATIONS COLLAPSE TO NEAR-CHANCE. METRIC: MACRO-F1 (SLEEP-EDFx), HEAD-ONLY ROC-AUC (TUAB). *TUAB IS A SEPARATE CONTROLLED REPROCESSING THROUGH OUR PIPELINE, SO ITS ABSOLUTE VALUES DIFFER FROM THE MAIN TUAB RESULT (TABLE V); THE 5–10 COMPARISON IS INTERNALLY CONTROLLED.

Dataset	Metric	1 s	5 s	10 s	20 s
Sleep-EDFx	macro-F1	0.048	0.539	–	0.060
TUAB*	ROC-AUC	–	0.771	0.776	–

b) Metric-aware adaptation matters under extreme imbalance: The RL head is decisive where the positive class is very rare: on the 0.64%-prevalence Siena corpus it lifts minority-class precision roughly eight-fold, and it yields the best TUSZ operating point. It is negligible where the classes are balanced (Sleep-EDFx), and—informatively—on the moderately imbalanced four-class TUEV task plain fine-tuning slightly exceeds it (macro-F1 0.530 versus 0.499). Practitioners should therefore expect a macro-F1-aware objective to matter most in the rare-positive regime rather than under mild imbalance, where additional fine-tuning capacity can suffice.

c) Temporal resolution bounds fine-grained tasks: The 5 s window with coarse pooling erases the spike-level morphology that separates epileptiform subtypes; merging the indistinguishable classes recovers a usable four-class TUEV task. Fine-grained temporal tasks will benefit from shorter windows or higher-rate probes.

d) The backbone is anchored to its 5 s pretraining scale: Because pretraining used a fixed five-second input, the backbone is most effective near that scale, and window length must be treated as a constraint rather than a free hyperparameter (Table XII). On Sleep-EDFx, macro-F1 peaks at 5 s (0.539) and collapses to near-chance at both 1 s (0.048) and 20 s (0.060): the 256-sample and 5120-sample inputs are far outside the 1280-sample pretraining regime, so the frozen features degrade. On the coarser TUAB abnormality task a modest extension to 10 s is tolerated and even helps slightly (head-only ROC-AUC 0.771→0.776; full-finetune accuracy 0.732→0.750), consistent with the 10 s window used by foundation-model benchmarks. The practical implication is that this fixed-length U-Net cannot ingest the long (30 s) epochs that sleep staging is scored on; matching such long-context tasks would require patch-based tokenization or a variable-length positional encoding, as in CBraMod and REVE, rather than a single fixed window.

Channel coverage is a further first-order factor: transfer quality broadly follows the number of real channels, and results on low-coverage datasets (Bonn, Sleep-EDFx) should be read as evidence of graceful degradation rather than as clean montage-transfer tests.

e) Clinical implications: Three properties align with the demands of clinical deployment. First, the extreme-imbalance behavior on Siena—an eight-fold precision gain and an order-of-magnitude reduction in false positives—directly addresses the dominant failure mode of continuous seizure monitoring, where a high false-alarm rate makes a detector unusable

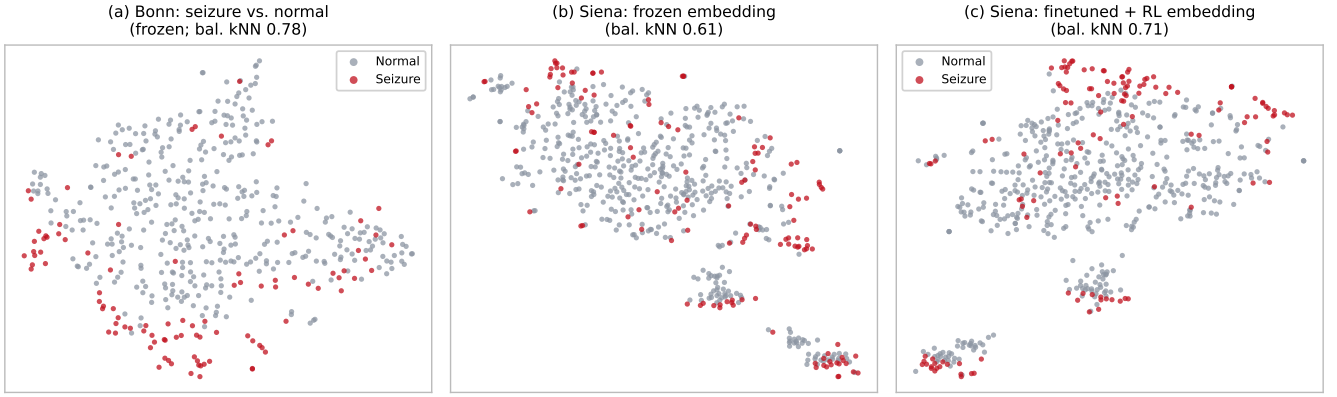


Fig. 5. t-SNE of the 4800-dimensional DiffEEG embedding, colored by class. (a) Bonn seizure vs. normal on the frozen pretrained embedding: the two classes separate without any task training. (b,c) Siena seizure vs. normal on the frozen and finetuned+RL embeddings: seizure windows form more coherent clusters after finetuning. Balanced k NN accuracy on the raw embedding is annotated per panel.

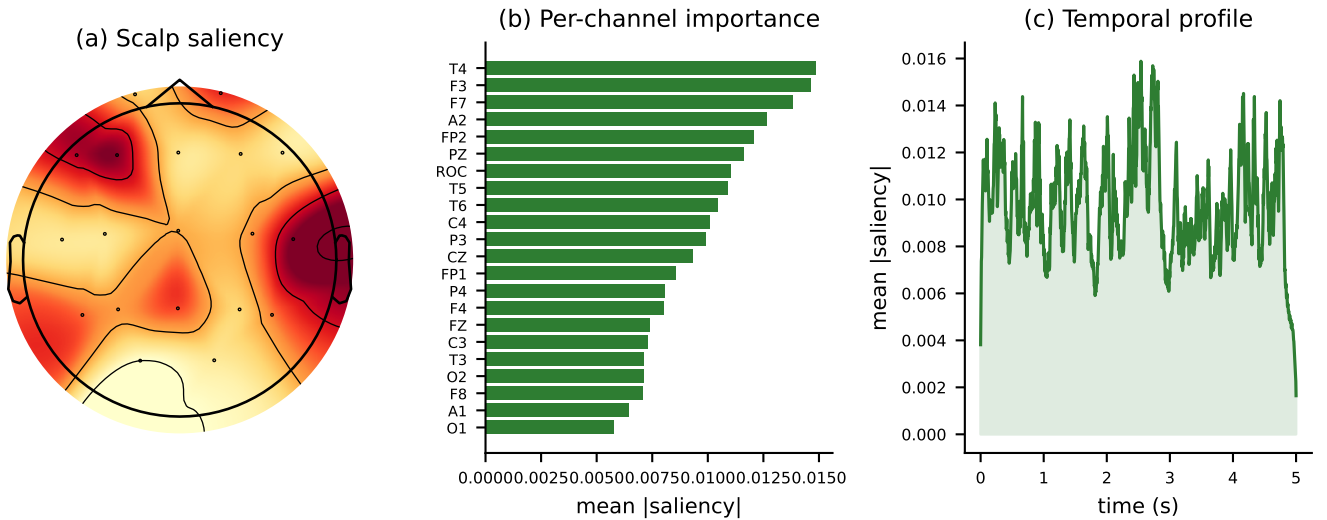


Fig. 6. Gradient saliency for the epileptiform class (TUEV). (a) Scalp topomap of per-channel importance over the 19 mappable 10–20 electrodes; hot regions are temporal/frontotemporal. (b) Per-channel importance ranking (all 22 model channels). (c) Temporal saliency profile across the 5 s window.

regardless of its AUROC. Second, the frozen embedding supports rapid, low-label adaptation: a new site or task can be served by training only a lightweight head on cached features, without the compute or data needed to retrain a foundation model. Third, the saliency and embedding analyses provide a physiologically plausible, inspectable rationale for each decision (Figs. 5, 6), which is a prerequisite for clinical trust and regulatory review. We emphasize, however, that these results are retrospective and benchmark-based; prospective, multi-site validation with clinician-in-the-loop evaluation remains necessary before any deployment claim.

f) Limitations: The comparison to foundation models is positional rather than strictly like-for-like: while all methods use the official patient-disjoint splits, they differ in montage, sampling rate, and window length, and matching every confound would require reprocessing under each protocol. Pretraining used a single corpus (TUSZ) and a fixed montage, so the model has no built-in montage invariance and relies on the harmonization front end; low channel coverage (Bonn,

Sleep-EDFx) confounds task difficulty with montage transfer. Finally, we evaluate a single backbone scale, leaving scaling behavior for diffusion EEG embeddings open.

VIII. COMPARISON WITH STATE-OF-THE-ART

Table XIII compares DIFFEEG with recent EEG foundation models on three clinical benchmarks with established baselines: TUAB abnormality detection (accuracy, ROC-AUC, AUC-PR), TUEV event classification (accuracy, weighted-F1), and Mumtaz2016 MDD detection (balanced accuracy, ROC-AUC, AUC-PR). Foundation-model numbers are self-reported; the task-specific baselines did not evaluate on TUAB/TUEV in their original papers, so their scores are taken from the common EEG-FM benchmark (originated by BIOT and reproduced by CBraMod/REVE), as detailed in the table footnote. All methods—including ours—evaluate on the official patient-disjoint TUAB/TUEV test partitions, so the comparison is fair with respect to patient-independence. It remains indicative

rather than strictly like-for-like in other respects: the baselines use their own preprocessing (typically 16–20 bipolar channels at 200 Hz, and, for TUEV, all 112 491 fixed-window segments), whereas DIFEEG uses a 22-channel referential montage at 256 Hz and an event-windowed four-class TUEV task.

On TUAB abnormality detection, where preprocessing is closest, DIFEEG attains accuracy 0.793, ROC-AUC 0.878, and AUC-PR 0.877. On both AUC metrics it surpasses every task-specific (non-foundation) baseline—the best of which, ST-Transformer, reaches ROC-AUC 0.871 and AUC-PR 0.852—and is on par with the foundation models BIOT and EEGPT, while remaining below the strongest transformer FMs (LaBraM, CBraMod, REVE at 0.90–0.92). Models explicitly designed for montage generalization and trained on far larger corpora (REVE: 60 000 h / 92 datasets) lead on cross-montage transfer; DIFEEG instead shows that a single generative backbone with a lightweight harmonization front end already transfers usefully across seizure, abnormality, motor, and sleep tasks at a competitive parameter budget.

On Mumtaz2016 MDD detection, under the matched CBraMod/REVE preprocessing a frozen DIFEEG head reaches balanced accuracy 0.812, ROC-AUC 0.935, and AUC-PR 0.942. On these AUC metrics it approaches the strongest task-specific baselines (e.g. SPaRCNet ROC-AUC 0.978) while trailing the transformer foundation models (CBraMod and REVE at ≥ 0.99)—the same ordering seen on TUAB, and again without any fine-tuning.

A. The effect of test-set class balance

A subtle but important confound underlies seizure-detection benchmarks. Foundation-model results on the TUH Seizure corpus are typically reported on a class-balanced test set, built by keeping seizure-containing records and undersampling background so that seizures form $\approx 29\%$ of the windows. Continuous clinical monitoring, by contrast, is naturally imbalanced: seizures occupy only $\approx 6.7\%$ of our patient-wise evaluation split. Because AUPRC depends strongly on prevalence while AUROC does not, evaluating on a balanced test set inflates AUPRC and makes a method appear far more precise than it would be in deployment.

To quantify this, we re-score the identical DIFEEG probabilities after subsampling the negative class to several target prevalences (200 random seeds; Table XIV). AUROC is invariant (≈ 0.836 frozen, ≈ 0.841 unfrozen) across all constructions, as expected. AUPRC, however, climbs steeply with prevalence: the unfrozen model rises from 0.484 at the natural 6.7% to 0.762 at the 29% construction used by seizure-FM benchmarks and to 0.862 at full balance, with the model completely unchanged. Evaluated at the same 29% construction as the baselines, DIFEEG (0.762) exceeds EEGWNet (0.74), is on par with MTP-Virgin (0.78), and trails the pretraining-heavy BioSerenity-E1 (0.85). Most of the apparent AUPRC gap is therefore an artifact of test-set balance rather than of representational quality; the honest, prevalence-invariant gap is in AUROC (≈ 0.84 vs. 0.926), which we attribute partly to the reference’s longer analysis windows

(16 s vs. our 5 s). On the companion TUH abnormality task, the same low-data study reports AUPRC 0.907 (BioSerenity-E1), 0.898 (EEGWNet), and 0.88 (MTP-Virgin); DIFEEG reaches 0.877 (Table V), competitive with these despite far less pretraining data.

IX. CONCLUSION

We presented DIFEEG, a denoising-diffusion foundation model whose frozen encoder provides a compact, transferable EEG representation. A multi-timestep diffusion-probing embedding, combined with a channel-harmonization front end and metric-aware reinforcement-learning adaptation, transfers across seven datasets and four task families—reaching 0.993 ROC-AUC on Bonn seizure detection, 94.0% accuracy on TUSZ seizure detection, 0.878 ROC-AUC on TUAB abnormality detection, and above-chance accuracy on motor imagery and sleep staging that the backbone never encountered during pretraining. Compared with recent foundation models, DIFEEG is competitive on clinical abnormality detection while offering broad, low-cost transfer from a single generative backbone. Future work will pursue montage-invariant positional encodings and multi-corpus diffusion pretraining to close the remaining gap to the largest foundation models.

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TABLE XIII

COMPARISON WITH TASK-SPECIFIC AND FOUNDATION EEG MODELS ON TUAB (ABNORMALITY), TUEV (EVENTS), AND MUMTAZ2016 (MDD DETECTION). VALUES ARE BALANCED ACCURACY (ACC), ROC-AUC, AND AUC-PR FOR TUAB; BALANCED ACCURACY AND WEIGHTED-F1 (W-F1) FOR TUEV; AND BALANCED ACCURACY (ACC), ROC-AUC, AND AUC-PR FOR MUMTAZ2016. FOUNDATION-MODEL ROWS ARE SELF-REPORTED; THE TASK-SPECIFIC BASELINES DID NOT ORIGINALLY EVALUATE ON THESE CORPORA, SO THEIR NUMBERS ARE TAKEN FROM THE UNIFIED EEG-FM BENCHMARK (SEE FOOTNOTE). BEST PER COLUMN IN BOLD; UNDERLINE MARKS THE BEST NON-FOUNDATION BASELINE.

Model	Params (M)	TUAB			TUEV		Mumtaz2016		
		Acc	ROC-AUC	AUC-PR	Acc	W-F1	Acc	ROC-AUC	AUC-PR
Task-specific (non-foundation)									
EEGNet [16]	0.003	0.764	0.841	0.830	0.388	0.654	0.923	0.964	0.963
EEGConformer [17]	0.55	0.776	0.845	0.843	0.407	0.698	0.931	0.970	0.968
SPaRCNet [18]	0.79	0.790	0.868	0.841	0.416	<u>0.702</u>	<u>0.932</u>	<u>0.978</u>	0.975
ContraWR [19]	1.6	0.775	0.846	0.842	<u>0.438</u>	0.689	0.920	0.962	0.959
CNN-Transformer [20]	3.2	0.778	0.846	0.843	0.409	0.685	0.931	0.974	<u>0.976</u>
FFCL [21]	2.4	0.785	0.857	0.845	0.398	0.678	0.931	0.975	0.972
ST-Transformer [22]	3.5	<u>0.797</u>	<u>0.871</u>	<u>0.852</u>	0.398	0.682	0.914	0.959	0.958
Foundation models									
BIOT [3]	3.3	0.796	0.881	0.879	0.528	0.749	0.936	0.976	0.974
EEGPT [6]	25.0	0.798	0.872	—	0.623	—	—	—	—
LaBraM-Base [4]	5.8	0.814	0.902	0.897	0.641	0.831	0.941	0.978	0.980
CBraMod [5]	4.2	0.829	0.923	0.926	0.667	0.834	0.956	0.992	0.992
NeurIPT [13]	73.5	0.829	0.895	0.904	0.676	—	—	—	—
REVE-Base [14]	69	0.832	0.925	0.928	0.676	0.845	0.964	0.996	0.996
BioSerenity-E1 [15]	11.7	0.823	0.890	0.910	—	—	—	—	—
DiffEEG (ours)	40.7	0.793	0.878	0.877	0.628 [†]	0.651 [†]	0.812 [‡]	0.935 [‡]	0.942 [‡]

Source of numbers: the task-specific baselines (EEGNet–ST-Transformer) did not evaluate on TUAB/TUEV in their original papers; their values are as re-implemented and reported in the unified EEG-FM benchmark introduced by BIOT [3] and reproduced by CBraMod [5] and REVE [14] (we cite the REVE appendix tables). The first-column citations identify the original methods, not the source of these TUAB/TUEV scores. Foundation-model rows (BIOT–BioSerenity-E1) and our own row are self-reported. All methods use the official patient-disjoint TUAB/TUEV train/test partitions (a single split, with validation drawn from the training subjects), so every test column is patient-independent; our TUAB uses the same official split. AUC-PR is a binary metric and is not defined for the multiclass TUEV task, which is therefore compared on accuracy and weighted-F1 (W-F1). EEGPT does not report TUAB

AUC-PR; EEGPT and NeurIPT report TUEV Cohen’s Kappa but not weighted-F1, hence “—”. The Mumtaz2016 baseline columns are the balanced-accuracy/AUROC/AUC-PR values from the REVE benchmark (its Table 9, subject-independent split, all EC+EO windows) [14]. EEGPT, NeurIPT, and BioSerenity-E1 did not evaluate on Mumtaz2016. [†] Four-class merged, event-windowed TUEV (macro-F1 0.530); not directly comparable to the baselines’ six-class setup. [‡] Frozen-head result under the CBraMod/REVE-matched preprocessing (EC+EO, 50 Hz notch + band-pass + common-average reference; Table IX).

TABLE XIV

SEIZURE-DETECTION AUPRC ON THE TUH SEIZURE CORPUS. DIFFEEG PROBABILITIES ARE RE-SCORED AT THREE SEIZURE PREVALENCES (200 SEEDS); AUROC IS PREVALENCE-INVARIANT, AUPRC RISES WITH THE SEIZURE FRACTION. BASELINE MODELS ARE AS REPORTED IN THE BIOSERENITY-E1 LOW-DATA STUDY [15] (BEST OF THE FINE-TUNING-HOURS CURVE) AT THEIR BALANCED ($\approx 29\%$) EVALUATION. THE 29% DIFFEEG ROW IS THE MATCHED COMPARISON POINT.

Model	Test prevalence	AUROC	AUPRC
DiffEEG frozen	6.7% (natural)	0.836	0.369
	29% (matched)	0.836	0.712
	49% (balanced)	0.836	0.838
DiffEEG unfrozen	6.7% (natural)	0.841	0.484
	29% (matched)	0.841	0.762
	49% (balanced)	0.841	0.862
EEGNet [15]	$\approx 29\%$	—	0.74
MTP-Virgin [15]	$\approx 29\%$	—	0.78
BioSerenity-E1 [15]	$\approx 29\%$	0.926	0.85

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